

# Luwei (Luis) Lin

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Edmonton, AB, Canada · Canadian citizen · No sponsorship needed · Open to relocate to the US

Full-stack software engineer with 2+ years shipping production features across SaaS platforms. Built ML pipelines (SVC, NER, TF-IDF) in Python and full-stack systems in Java, React/TypeScript, backed by PostgreSQL and MongoDB. Experienced with cloud infrastructure on AWS using Docker and CI/CD. Holds CCIE #65030 with deep TCP/IP and distributed systems fundamentals.

## EDUCATION

### University of Illinois Urbana-Champaign — Master of Computer Science

Sept 2026 – Dec 2027 (Expected)

Urbana-Champaign, IL, USA

### University of Alberta — B.Sc. Computing Science & Statistics

Sept 2021 – Apr 2026

Edmonton, AB, Canada

## PROFESSIONAL EXPERIENCE

### Software Engineer, Intern — Arista Networks, Routing Team

Sept 2023 – Dec 2023

Vancouver, BC · Linux, C++, Python3, EOS, YAML

- Reduced BGP/OSPF route convergence time on Arista EOS datacenter switches by redesigning graph traversal algorithms (DFS/BFS) in C++, directly improving network recovery performance in production environments.
- Diagnosed and resolved critical routing protocol bugs in large-scale datacenter network environments; implemented structured error logging to detect and isolate anomalous routing behavior across fleet deployments.
- Designed test plans and automated validation frameworks in Python3 for large-scale routing feature verification across datacenter network topologies.
- Automated router configuration provisioning for cloud datacenter deployments using YAML-based infrastructure-as-code, enabling scalable and repeatable network rollouts.

### Full-Stack Software Engineer — Avanti Software

May 2024 – Present

Remote, Edmonton AB · Java, React, TypeScript, ASP.NET, PostgreSQL, Redis, Docker, AWS, C#

- Architected and deployed a scalable full-stack payroll system (Java, React, TypeScript) for 300+ enterprise clients, supporting critical financial workflows with strong uptime and reliability.
- Engineered a distributed, event-driven batch processing pipeline to manage highly concurrent state mutations, using fault tolerance and idempotency to ensure reliable execution.
- Optimized complex relational schemas and query plans in PostgreSQL, then integrated distributed caching with Redis to resolve high-concurrency bottlenecks.
- Spearheaded cloud infrastructure automation and containerization with AWS and Docker, designing robust CI/CD pipelines for faster, safer production rollouts.

### Backend ML Developer, Contract — ShopHopper

Sep 2022 – Apr 2023

Kelowna, BC · Python3, PostgreSQL, scikit-learn

- Built an ML pipeline using Linear SVC to classify and decode product strings for cross-catalog matching.
- Implemented a recommendation engine that matched user preferences to products and improved purchase conversion.

### Data Analyst, Intern — Mitacs

May 2022 – Aug 2022

Edmonton, AB · Python3, NLP, spaCy, Jupyter

- Trained a NER model in Python/spaCy to extract structured product attributes such as size, color, and pattern from raw text at scale.
- Built a text normalization pipeline with TF-IDF and n-gram phrase matching to sanitize and standardize 30K+ product records.

## RELEVANT PROJECTS

### Real-Time Video Call & Chat App

2021

- Built a full-stack real-time communication app using WebRTC, ICE/TURN/STUN protocols, and Node.js.

### Wildfire Prediction — Borealis AI

Oct–Nov 2021

- Processed 2M weather records, implemented LSTM and regression models, and built a web crawler that generated a 1M-record training dataset.

## TECHNICAL SKILLS

|                |                                                            |
|----------------|------------------------------------------------------------|
| Languages      | Python3, TypeScript, C#, JavaScript, C++, C, Java, R       |
| Frameworks     | React, Next.js, ASP.NET, Node.js, Flask, WebRTC, Socket.io |
| AI / ML        | scikit-learn, spaCy, LSTM, TF-IDF, NER, SVC, Pandas, NumPy |
| Databases      | PostgreSQL, MongoDB, MySQL, SQLite3, Redis                 |
| Infrastructure | AWS, Docker, Linux/Unix, Git, Jenkins, CI/CD, YAML         |

## CERTIFICATIONS

CCIE Routing & Switching #65030

CCNP-RS

CCNA-RS

Google Data Analytics

Dean's Honor Roll